

UNIVERSITY OF HELSINKI

Project work: Search for the Higgs Boson in a Generator-Level Analysis

Computing Methods in High Energy Physics, PAP331

Siiri Hilska, Shounak Naik

22.05.2026

Contents

1	Generating the Signal and the Background	2
2	Analysis on the Generated Data	2
a	Identifying the signal Muons	2
b	Reconstructing the invariant mass $M_{4\mu}$	6
c	Fitting and estimating the statistical significance	6
3	On the Validity of our Results	8

1 Generating the Signal and the Background

The signal and background samples were generated using PYTHIA 8, a Monte Carlo event generator for high-energy particle collisions. Both samples simulate proton-proton collisions at a centre-of-mass energy of $\sqrt{s} = 13.6$ TeV, corresponding to LHC Run 3 conditions. The signal process is the Standard Model Higgs boson produced via all available production channels (`HiggsSM:all = on`). The Higgs boson (PDG ID 25) was forced to decay exclusively to two Z bosons (PDG ID 23) (`25:onIfMatch = 23 23`), and each Z boson was forced to decay exclusively to a muon-antimuon pair (`23:onIfMatch = 13 -13`), yielding the golden channel $H \rightarrow ZZ^* \rightarrow 4\mu$. The background process is continuum diboson production via $q\bar{q} \rightarrow \gamma^*/Z \gamma^*/Z$ (`WeakDoubleBoson:ffbar2gmZgmZ = on`), with the full γ^*/Z interference included via `WeakZ0:gmZmode = 0`. As with the signal, both vector bosons were forced to decay to $\mu^+\mu^-$ pairs, producing an irreducible $ZZ/\gamma^* \rightarrow 4\mu$ background. Both samples used a time-based random seed to ensure statistically independent generations.

A total of 10^6 events were generated, split between the two samples according to their cross-section ratio: approximately 946 938 signal events and 53 062 background events. For each event, all final-state particles were looped over. Muons (PDG ID ± 13) and charged pions (PDG ID ± 211) were identified and their kinematic quantities: transverse momentum p_T , pseudorapidity η , azimuthal angle ϕ , and charge q , were stored as ROOT vector branches in a TTree called **Events**. Pion information was saved alongside muon information to facilitate the track isolation step in the analysis. A soft dimuon trigger was applied at generation level, requiring at least two muons with $p_T > 20$ GeV and $|\eta| < 2.1$. Only events passing this trigger were stored. A separate **Meta** tree was also written to each ROOT file, storing the total number of generated events, the number passing the trigger, and the generator-level cross section σ and its uncertainty as reported by PYTHIA, which are used for event weight normalisation.

2 Analysis on the Generated Data

a Identifying the signal Muons

In identifying the signal muons, the following standard selection criteria were implemented:

1. First, the muon measurement error was simulated by adding a 1% Gaussian smearing to their momenta and a 2 mrad Gaussian smearing to their angles θ and ϕ .
2. Then, it was required that the smeared muons satisfy $p_T > 30$ GeV/c.
3. Lastly, it was required for charged pions within $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < 0.3$ of a muon candidate that the sum of momenta $\Sigma p_T < 1.5$ GeV/c. After this step at least four muon candidates had to exist in the event.

Here's an explanation on how this was actually done in the code:

Smearing:

The ROOT file filled with the generated signal and background values was opened. From the tree in the file the branch containing p_T , η , ϕ values called "Event" was accessed. Then, a new ROOT file with a new tree was created for storing the smeared values. For the smeared muon values ROOT vector objects were created and for each a branch of their own to the new tree. Also, into the same tree branches and vectors were created for storing pion values for the third step of the selection criteria.

A pseudorandom number generator was initialized with the seed being the current timestamp. Then, the original tree was looped over. From each entry original muon p_T , η , ϕ and q value was taken. From η θ was calculated with $\theta = 2 \arctan(e^{-\eta})$. After that, the pseudorandom number

generator was used for adding the desired 1% and 2 mrad Gaussian smearing to p_T , θ and ϕ . Lastly, the smeared values were stored into the new tree.

The same procedure was repeated for pions, except that their values weren't smeared. Instead, the original pion values were stored into the new ROOT tree into branches of their own. This was done to make it easier to access the pion values while performing the third step of the selection criteria.

After each event looped from the original tree, the vectors were emptied to avoid any mismatches. When the whole original tree had been looped over and the muon and pion values filled to the new tree, the whole tree was written into a new file with the smeared muon values and original pion values.

To check the smearing had worked, histograms with the original and smeared p_T , θ and ϕ values were created. In each histogram the original values are drawn in blue while the smeared values are red. The smearing histograms can be seen below.

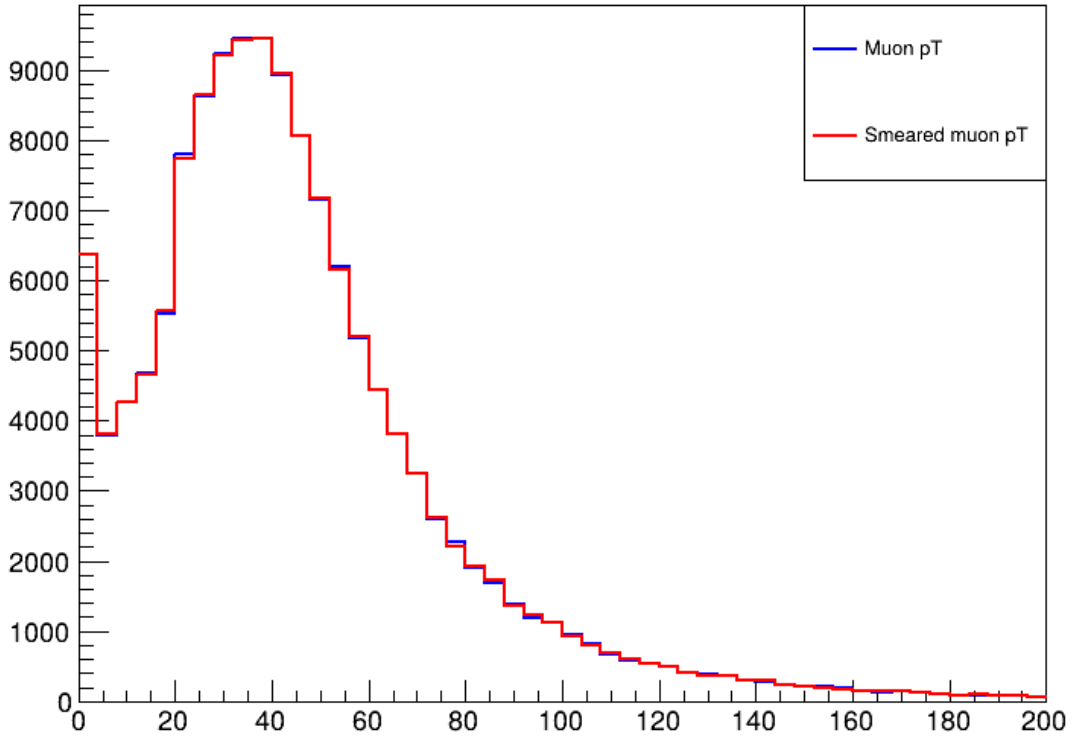


Figure 1: The p_T smearing histogram

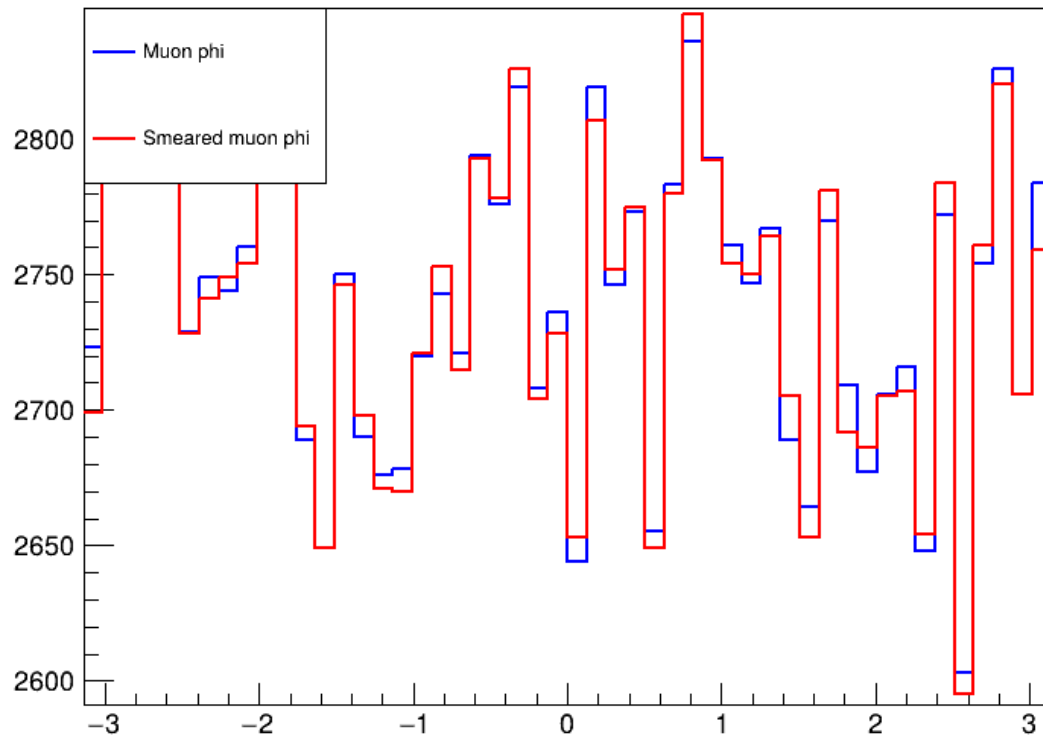


Figure 2: The phi smearing histogram

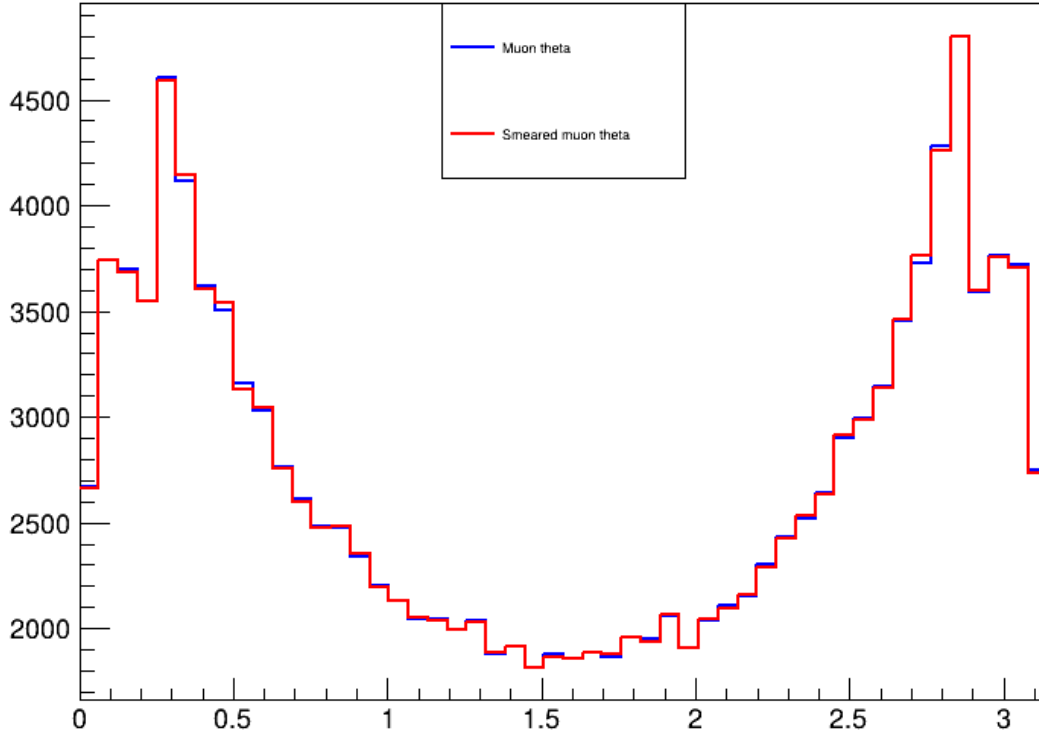


Figure 3: The theta smearing histogram

Momentum requirement and track isolation:

The new ROOT file with the smeared values was opened and the tree within the file accessed. Each entry (event) from the tree was looped over. A variable was initialized to keep track of the amount of muons passing the track isolation criterion.

For each event muons were looped over. First, it was checked whether the muon meets the momentum requirement of $p_T > 30$ GeV/c. If not, the muon was just skipped. If the requirement was satisfied, the code moved on to the next part in which the pions of the event are looped over. Before starting the loop, a variable was initialized to keep track of the total momenta of the pions in the event. For each pion ΔR is calculated using ROOT's own DeltaR function. For the function to be used, the muon and pion values have to be turned into TLorentzVector four-vector objects with p_T , η , ϕ and the mass of the particle. When ΔR has been calculated the code checks whether the pion is within the desired $\Delta R < 0.3$. If the pion is within the range, its momentum is summed into the variable keeping count of the total sum of the momenta of charged pions.

After all pions of an event have been summed, it is checked whether they fulfill the condition of the sum of the momenta being less than 1.5 GeV/c. Again, if the condition isn't fulfilled the rest of the loop is skipped. If instead the requirement is met the muons are added into a list as a four-vector object containing the p_T , η , ϕ and mass values of the muon. Lastly, it is checked whether at least four muons still exist in the event and if it's true 1 is added to the variable keeping track of the muons passing track isolation.

From the total number of 946937 signal events 37600 events passed this selection. From the total number of 53062 background events 612 events passed this selection.

b Reconstructing the invariant mass $M_{4\mu}$

The invariant mass of the four-muon system was reconstructed from the smeared signal and background ROOT files produced in Part 2a. Two histograms were created, one each for signal and background, with 50 bins spanning 50-180 GeV. Since the number of generated events does not in general equal the number of events expected at a given integrated luminosity, each histogram entry was weighted by

$$w = \frac{\sigma_{\text{gen}} \cdot \mathcal{L}}{N_{\text{total}}}, \quad (1)$$

where σ_{gen} is the generator-level cross section in fb read from the meta tree, $\mathcal{L} = 110 \text{ fb}^{-1}$ is the target integrated luminosity, and N_{total} is the total number of generated events before any selection. For each event the smeared muon candidates were looped over. Muons with smeared $p_T \geq 30 \text{ GeV}$ were required. The θ value was limited to $(10^{-6}, \pi - 10^{-6})$ to avoid numerical divergences in the η conversion. The track isolation criterion from Part 2a was then applied: for each muon candidate a TLorentzVector was constructed using its smeared p_T , η , and ϕ with the muon mass $m_\mu = 105.66 \text{ MeV}$, and similarly for each charged pion using the pion mass $m_\pi = 139 \text{ MeV}$. The angular separation ΔR between the muon and each pion was computed using ROOT's built-in method. If the scalar sum of p_T of all pions within $\Delta R < 0.3$ exceeded 1.5 GeV, the muon was rejected as non-isolated. Events with fewer than four muon candidates surviving these requirements were discarded. For events with at least four selected muons, the four leading muon candidates were combined by summing their TLorentzVector objects. The invariant mass of the resulting four-vector,

$$M_{4\mu} = \sqrt{\left(\sum_{i=1}^4 p_i\right)^2}, \quad (2)$$

was computed via ROOT's TLorentzVector::M() method and filled into the appropriate histogram with the event weight w .

c Fitting and estimating the statistical significance

A couple of things were done to complete the data analysis. First, a function was fitted to the signal+background $M_{4\mu}$ histogram. For the function, we chose a mixture of a Breit-Wigner and a decreasing exponential. The Breit-Wigner distribution function was chosen to account for the signal effects. The function was available directly from ROOT and is of the following form:

$$f(E) = \frac{k}{(E^2 - M^2)^2 + M^2\Gamma^2} \quad (3)$$

$$\text{where } k = \frac{2\sqrt{2}M\Gamma\gamma}{\pi\sqrt{M^2 + \gamma}} \quad (4)$$

$$\gamma = \sqrt{M^2(M^2 + \Gamma^2)} \quad (5)$$

Here E is the center-of-mass energy that produces the resonance, M is the mass of the resonance and Γ is the resonance width.

The decreasing exponential was chosen to fit the overall decreasing trend of the histogram. Some sensible initial parameters were chosen by trial and error and some limits were set to keep the fit contained. After fitting the full signal+background histogram the background was fitted separately with a decreasing exponential only. This fit was drawn with a blue dashed line to make it distinct from the red line of the overall fit. Also, a Breit-Wigner function was chosen for the signal for later analysis even though the fit wasn't added to the canvas.

After the fits were done a reasonable mass window was chosen for calculating the number of signal and background events. As the mass window we chose for the lower limit the mean of

the mass minus 2σ and for the upper limit mean of the mass plus 2σ . The fit functions were integrated over within the chosen mass window. The calculated integrals were divided by the bin width of a single histogram entry to actually get the number of events. Then, the statistical significance was estimated with a naive expression $\alpha = N_s/\sqrt{N_B}$ where N_s is the number of signal events and N_b the number of background events. For a discovery we would need a 5σ signal, so next we calculated in the code how much integrated luminosity we would need for such a signal with the formula $\mathcal{L}_{\text{needed}} = \mathcal{L}_0(\frac{5}{\alpha})^2$, where $\mathcal{L}_0$ is the luminosity of our current data set.

At last, we approximated how long it would take to reach such an integrated luminosity in CMS in the Run3 data taking. We checked online how much luminosity $\mathcal{L}_{\text{Run3}}$ Run3 delivers yearly from their public results website and found that as of now CMS Run3 records 326.03 fb^{-1} yearly (see figure 5). With that info the time needed was calculated as $t = \frac{\mathcal{L}_{\text{needed}}}{\mathcal{L}_{\text{Run3/year}}}$. And thus the time needed would be approximately 760 days (or 2.08 years).

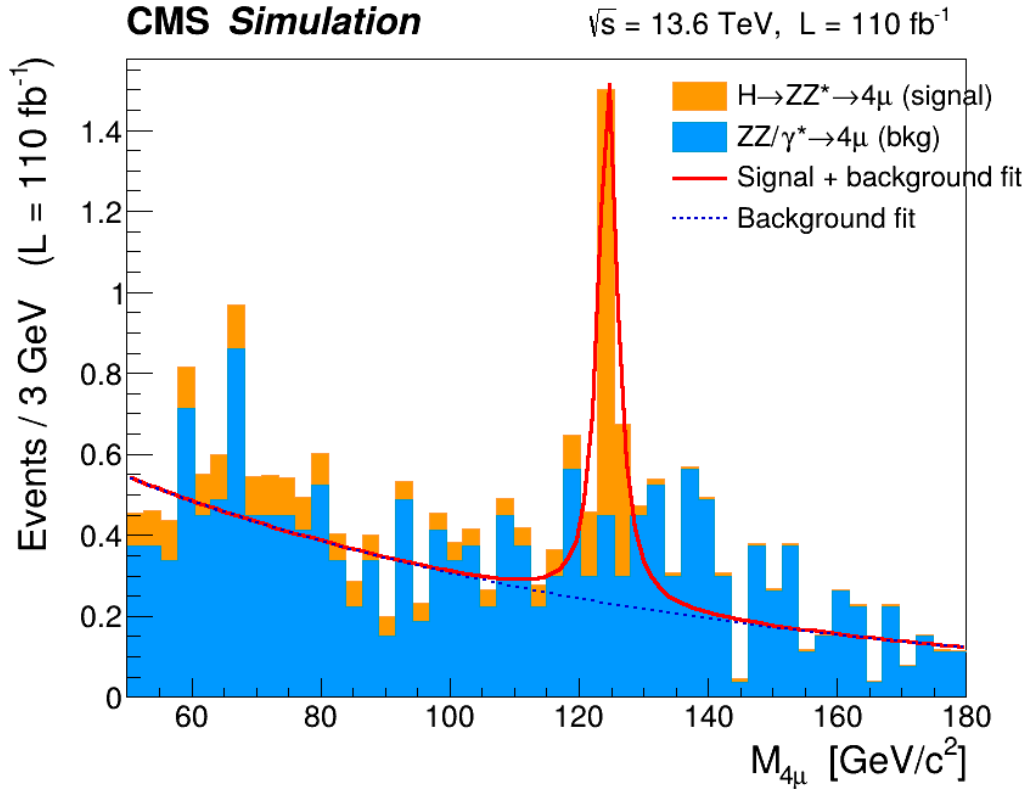


Figure 4: The invariant mass spectrum with the total fit drawn in red and the background fit drawn in a blue dashed line.

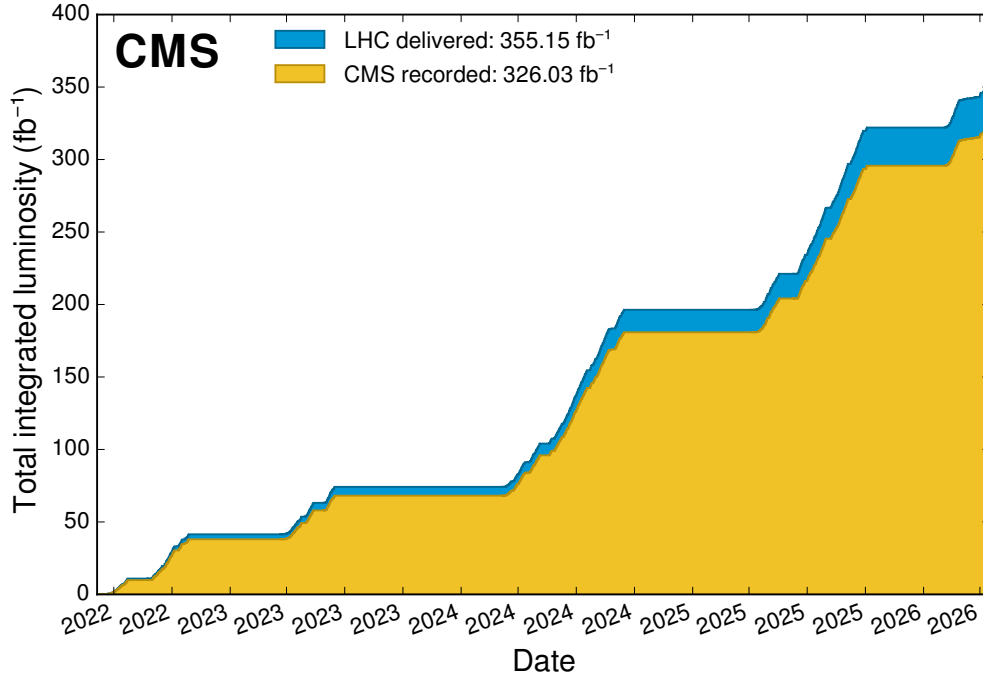


Figure 5: The integrated luminosity produced by CMS Run3 as reported in the public results website.

3 On the Validity of our Results

Even though our analysis utilizes procedures used in real particle physics measurements and analyses, one should keep in mind that it is somewhat a minimal and simplified exercise. In our results we observe a visible peak above the background in the invariant mass spectrum. However, the background we used is simplified and may not accurately represent what one might find in a real measurement.

In the track isolation part we only checked for charged pions whether they are within $\Delta R < 0.3$. The next part would be to expand the step to check for other particles within the zone.

The code should be run multiple times on a longer time scale and different fits should be tried. It could be beneficial to also try different event generators. Detector effects should be studied and taken into account further. For example, one could run a full detector simulation with GEANT4.

In a simulation it is difficult to check whether there could be a systematic error affecting our results. In a real study one could find for example detector calibration issues. Currently, our code only takes into account the statistical uncertainties. We also used the naive expression $\alpha = N_s / \sqrt{N_B}$ for the statistical significance. More sophisticated statistical methods could be used.

Still, the methodology used is transparent and reproducible with PYTHIA8 and ROOT, making our analysis credible as a baseline study. We implemented realistic selection criteria, included smearing to illustrate detector effects and reconstructed the signal in a clear way showing that the signal peak is visible under LHC Run3 conditions.